Networks and Distributed Programming

Reti e Programmazione Distribuita

Università di Napoli Federico II – Scuola Politecnica e delle Scienze di Base
Corso di Laurea in Informatica

Alberto Finzi

(alberto.finzi@unina.it)





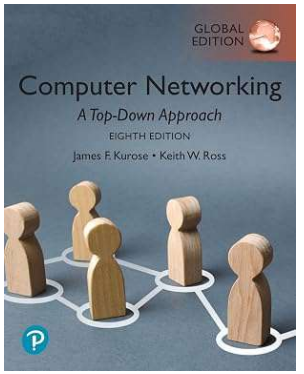
Overview

About the Course

- **Network and Distribuited Programming (Reti e Programmazione Distribuita):**
  - CFU: 9
  - Modality: Frontal lessons and class examples (in Italian).
  - Team code: **rpmzm7k**
    - Communications will be mostly provided on Teams

- **Evaluation:**
  - Written and oral examination.
  - Dates depend on the calendar.

- **Text books:**
  - **James Kurose, Keith Ross, Computer Networking A Top-Down Approach.** [Main reference]
  - Andrew Tanenbaum, David Wetherall, Computer Networks.
  - W. Richard Stevens, Unix Network Programming



- **Additional Materials:**
  - Slides (in English).

Overview

Contacts

- Alberto Finzi
  - Dipartimento di Ingegneria Elettrica e Tecnologie dell'Informazione (DIETI)
    - **E-mail**: alberto.finzi@unina.it
    - **Teams:** alberto.finzi@unina.it
    - **Web (UNINA):** https://www.docenti.unina.it/alberto.finzi
    - **Workplaces:**
      - Via Claudio 21, building 1
      - Via Claudio 21, building 5/A, PRISMA Lab

Overview
Schedule and Dates

- Class Schedule:

Giorno	Ora	Luogo
Martedì	12:30-14:30	Via Claudio 21, aula CL-I-1
Giovedì	14:30-16:30	Via Claudio 21, aula CL-I-3
Venerdì	14:30-16:30	Via Claudio 21, aula CL-T-2

- Assistantship (Ricevimento):

Giorno	Ora	Luogo
Martedì	14:30 – 16:30	Teams - Via Claudio 21, edificio 1
	Teams	On appointment

Overview

Motivations

- **We live in a connected world.**
  - 6.12 billion people use the Internet, 73.8% of the world population (*yet about 2.2 billion people are still offline)* [GWI/DataReportal 2026]
  - The typical Internet user spends more than 6½ hours online per day [GWI/DataReportal 2025].



- **Most modern software is networked software.**
  - Web, cloud, streaming, messaging, online games, AI services, IoT, …

- **Information and Communication Technologies (ICT)**
  - Communication is a core component of modern computing

Overview

Objectives

- Introduce fundamental elements on which **computer networks** (in general) and **internet** (in particular) are based.
- Provide **basic concepts and tools** allowing students to understand, analyze, design, and develop **computer networks and network applications**.
- Introduce and understand **network protocols**, having knowledge about **networks infrastructure and devices**.
- Provide basic concepts and tools to design and develop **concurrent and distributed network applications**.
- To understand potentialities, as well as **risks, limitations and security** issues of computer networks.

Overview

Course Perspective

- Two complementary perspectives:
  - **Computer Networking** and **Concurrent/Distributed Programming**

- **Networking:** how computers communicate protocols, architectures, network services

- **Programming:** how to build software that communicates; processes, threads, sockets, concurrent servers

- **Goal:** understand both the **communication infrastructure** and the **software that uses it**

Overview

Syllabus

Network and Distributed Programming — Networking

- *Introduction to computer networks:* goals and applications, history of the Internet, main concepts, network topologies and types, services.

- *Layered models:* ISO/OSI reference model and TCP/IP architecture.
- *Application layer:* client-server and peer-to-peer architectures, FTP and SSH protocols, structure of URL, HTTP protocol, e-mail protocols, DNS, BitTorrent.
- *Transport layer:* fundamentals, UDP and TCP, reliability, flow and congestion control.
- *Network layer:* addressing and forwarding, routers, IPv4 and IPv6, DHCP, NAT, routing fundamentals and algorithms.
- *Access layer* (data link + physical): network interfaces, parity checking and CRC, collisions, ARP protocol, Ethernet protocol (basis).

- *Introduction to network security:* attacks on networks, introduction to cryptography, message integrity, operational security.

Overview

Syllabus

Network and Distributed Programming — Distributed Programming

- *Concurrent programming:* processes and threads, concurrency models, communication and synchronization mechanisms, coordination of concurrent activities.
- *Distributed programming:* Network programming with socket and main APIs, UDP and TCP protocols.
- *Concurrent servers:* design and implementation of servers handling multiple clients.
- *Distributed applications:* communication and data exchange among networked services.